

CompTIA.

A+ Core 2

Exam 220-1202

A+ 220-1202

Module 22



Implementing Operational Procedures

Learning Outcomes

Prepare for A+ Core 2 by:

- Implementing best practice documentation to track assets
- Using common safety and environmental procedures
- Explaining basic scripting constructs and use cases

Lesson 22.1

Change and Inventory Management

Change Requests

Policies and procedures that reduce risk of outages caused by changes

Should include:

- Purpose, scope, type, timeline, effect

Risk Analysis



Identify possible issues with implementation



What controls or monitors can be implemented to reduce the risk



Identify cause and effect



Qualitative and quantitative analysis

Change Board Approvals

Change advisory board (CAB)

Officially approves changes after request is thoroughly reviewed



Some changes may not need full approval of board,
especially during incident response or emergency
procedures

Implement change

Rollback and/or backup plan

End-user acceptance

Asset Management

- Inventory
 - Name, make, model
 - Asset ID
 - Tag or barcodes are popular
 - Manufacturer
 - Specifications
 - Dates
 - Warranty, purchase, etc.
 - Cost
- Configuration management database (CMDB)
- Lifecycle
 - Procurement through disposal/destruction



Warranty and Licensing



Invoice documentation



Warranty timelines and paperwork



Document limits and license allocations/use

Lesson Review: Change and Inventory Management

- Change Requests
- Risk Analysis
- Change Board Approvals
- Implementation and Acceptance
- Asset Management
- Warranty and Licensing

Lesson 22.2

Common Safety and Environmental Procedures

Compliance with Regulations

Health and safety

Building codes

- Fire and electrical system protection

Environmental regulations

- Waste disposal like chemicals and lithium batteries

Electrical Safety (Slide 1 of 2)

- Current- amps
- Voltage- volts
- Resistance- ohms
- Tools
 - Voltage testers
 - Fuses
 - Grounding
- Power handling
- Electrical fire safety- CO2 or powder extinguishers vice water
- Equipment placement considerations

Electrical Safety (Slide 2 of 2)



Image © macmackyky 123RF.com



Image by phadventure © 123RF.com

Other Safety Hazard Mitigations

Trip hazards

Lifting techniques

- Legs not back
- Use multiple people when needed

Safety goggles and mask

Environmental Impacts

- Dust cleanup
 - Mask, vacuum, compressed air
- Temperature
- Humidity
- Ventilation



Image © 123RF.com

Electrostatic Discharge Mitigation (Slide 1 of 2)



Image by Audrius Merfeldas ©123RF.com.

- ESD can damage components
- ESD strap, mat, workbench
- Antistatic bag or dissipative packaging for component storage

Electrostatic Discharge Mitigation (Slide 2 of 2)

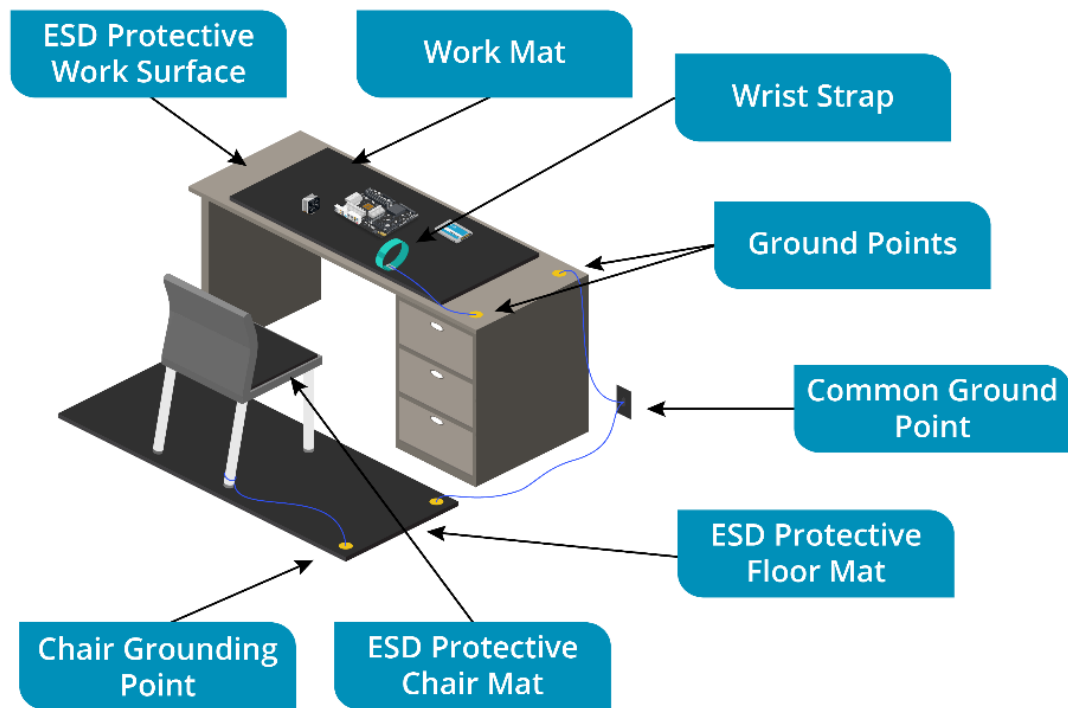


Image by Audrius Merfeldas ©123RF.com.

Building Power Issues and Mitigations (Slide 1 of 2)



Surge, under-voltage, failure



Surge suppressors



Battery backups or uninterruptable power supply (UPS)

Not meant for long term secondary power



Generator

Gas, Diesel, Propane, Natural gas

Building Power Issues and Mitigations (Slide 2 of 2)



Image by magraphics© 123RF.com.

Materials Handling and Responsible Disposal



Material safety data sheet (MSDS)

Hazards and cleanup/safety requirements



Proper disposal

Battery

Toner

Chemicals

Devices and assets

Lab Activity

- Lab: Install a UPS
(available in CertMaster Learn and CertMaster Perform)

Lesson Review: Common Safety and Environmental Procedures

- Compliance with Regulations
- Electrical Safety
- Other Safety Hazard Mitigations
- Environmental Impacts
- Electrostatic Discharge Mitigation
- Building Power Issues and Mitigations
- Materials Handling and Responsible Disposal

Lesson 22.3

Scripting Basics

Shell Scripts (Slide 1 of 2)

Specific to an OS

Script

- Series of commands that are executed in order

Integrated development environment (IDE)

Linux shell .sh

- Bash (Bourne Again Shell)
- Ksh (Korn Shell)

Shell Scripts (Slide 2 of 2)

```
1 #!/bin/bash
2 echo 'Hello World'
~
~
~
:set number                2,18      All
```

```
toor@LX20D:~$ chmod u+x hello.sh; ls -l $_
-rwxrw-r-- 1 toor toor 31 Jan 11 10:02 hello.sh
toor@LX20D:~$ ./hello.sh
Hello World
toor@LX20D:~$
```

Basic Script Constructs (Slide 1 of 2)

- Comments
 - Not read by compiler or interpreter
- Variables
- Branches
- Loops
- Operators

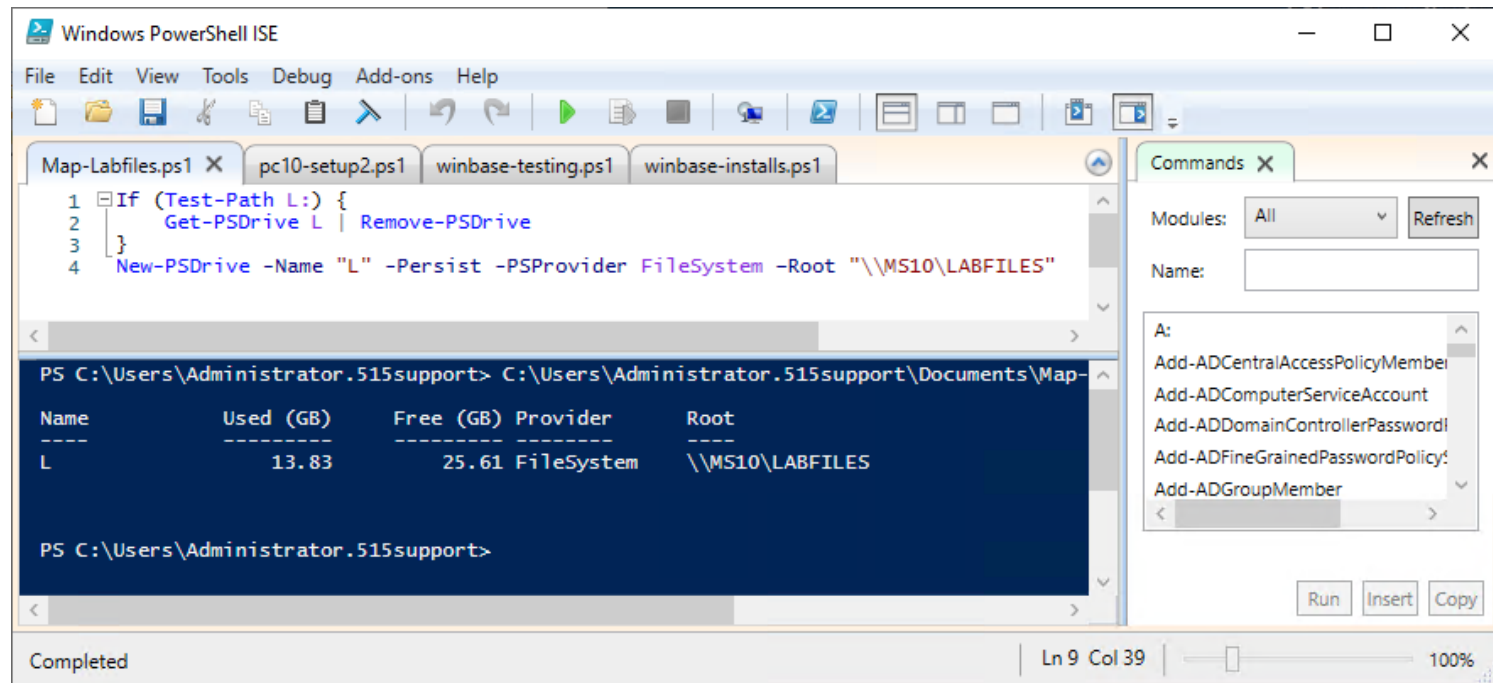
Basic Script Constructs (Slide 2 of 2)

Symbol Notation	Switch Notation	Usage
==	-eq	Is equal to (returns TRUE if both conditions are the same)
!=	-ne	Is not equal to (returns FALSE if both conditions are the same)
<	-lt	Is less than
>	-gt	Is greater than
<=	-le	Is less than or equal to
>=	-ge	Is greater than or equal to
&&	AND	If both conditions are TRUE, then the whole statement is TRUE
	OR	If either condition is TRUE, then the whole statement is TRUE

Windows Scripts (Slide 1 of 2)

- PowerShell .ps1
- VBScript .vbs
- Batch file .bat

Windows Scripts (Slide 2 of 2)

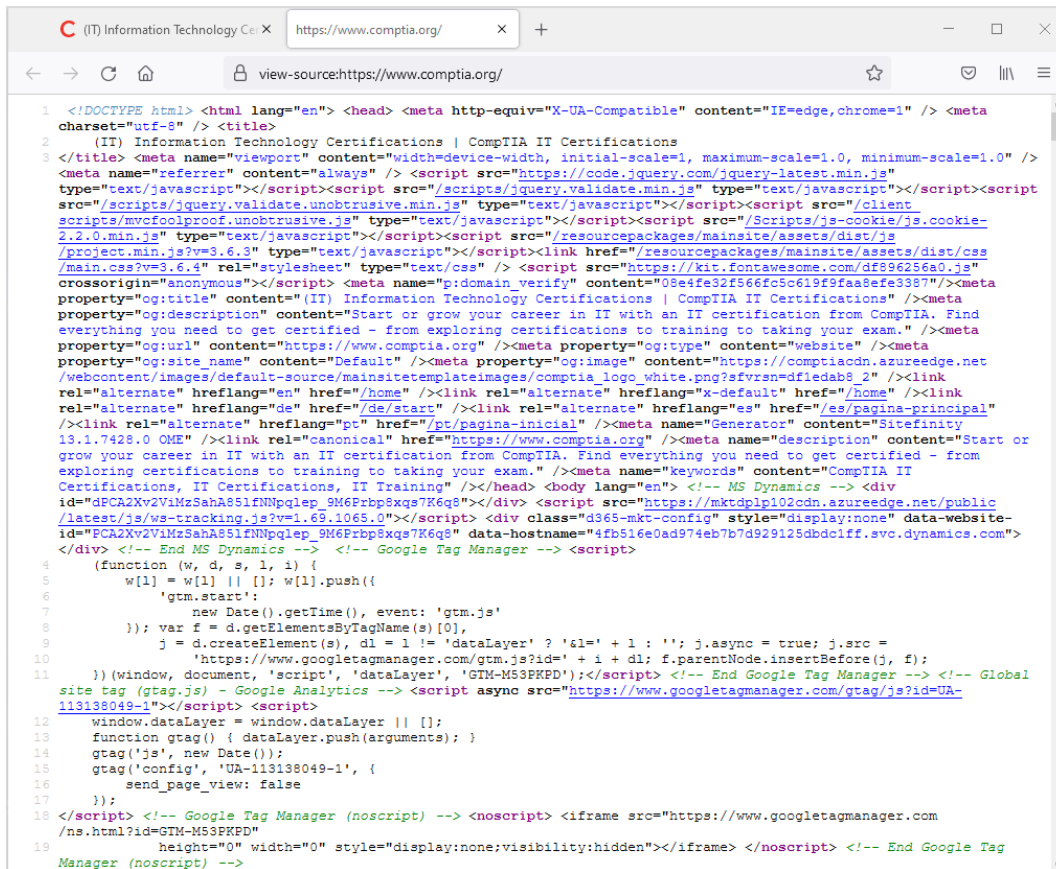


Screenshot courtesy of Microsoft.

JavaScript and Python (Slide 1 of 3)

- JavaScript .js
 - Web content and apps
- Python .py
 - General purpose primarily used in automation and software apps
 - Python IDLE

JavaScript and Python (Slide 2 of 3)



```
1 <!DOCTYPE html> <html lang="en"> <head> <meta http-equiv="X-UA-Compatible" content="IE=edge,chrome=1" /> <meta
2 charset="utf-8" /> <title>
3 (IT) Information Technology Certifications | CompTIA IT Certifications
4 </title> <meta name="viewport" content="width=device-width, initial-scale=1, maximum-scale=1.0, minimum-scale=1.0" />
5 <meta name="referrer" content="always" /> <script src="https://code.jquery.com/jquery-latest.min.js"
6 type="text/javascript"></script><script src="/scripts/jquery.validate.min.js" type="text/javascript"></script><script
7 src="/scripts/jquery.validate.unobtrusive.min.js" type="text/javascript"></script><script src="/client
8 scripts/mvcfoolproof.unobtrusive.js" type="text/javascript"></script><script src="/Scripts/js-cookie/js.cookie-
9 2.2.0.min.js" type="text/javascript"></script><script src="/resourcepackages/main/assets/dist/js
10 /project.min.js?v=3.6.3" type="text/javascript"></script><script src="/resourcepackages/main/assets/dist/js
11 /main.css?v=3.6.4" rel="stylesheet" type="text/css" /> <script src="https://kit.fontawesome.com/df896256a0.js"
12 crossorigin="anonymous"></script> <meta name="p:domain_verify" content="08e4fe32f566fc5c619f9faa8efe3387"/><meta
13 property="og:title" content="(IT) Information Technology Certifications | CompTIA IT Certifications" /><meta
14 property="og:description" content="Start or grow your career in IT with an IT certification from CompTIA. Find
15 everything you need to get certified - from exploring certifications to training to taking your exam." /><meta
16 property="og:url" content="https://www.comptia.org" /><meta property="og:type" content="website" /><meta
17 property="og:site_name" content="Default" /><meta property="og:image" content="https://comptiacdn.azureedge.net
18 /webcontent/images/default-source/mainstemplateimages/comptia_logo_white.png?sfvrsn=dfedab8_2" /><link
19 rel="alternate" hreflang="en" href="/home" /><link rel="alternate" hreflang="x-default" href="/home" /><link
20 rel="alternate" hreflang="de" href="/de/start" /><link rel="alternate" hreflang="es" href="/es/pagina-principal"
21 /><link rel="alternate" hreflang="pt" href="/pt/pagina-inicial" /><meta name="Generator" content="Sitefinity
22 13.1.7428.0 OME" /><link rel="canonical" href="https://www.comptia.org" /><meta name="description" content="Start or
23 grow your career in IT with an IT certification from CompTIA. Find everything you need to get certified - from
24 exploring certifications to training to taking your exam." /><meta name="keywords" content="CompTIA IT
25 Certifications, IT Certifications, IT Training" /></head> <body lang="en"> <!-- MS Dynamics --> <div
26 id="dPCA2Xv2ViMzSahA851fNnpqlep_9M6Prbp8xqs7K6q8"></div> <script src="https://mktcdplp102cdn.azureedge.net/public
27 /latest/js/ws-tracking.js?v=1.69.1065.0"></script> <div class="d365-mkt-config" style="display:none" data-website-
28 id="PCA2Xv2ViMzSahA851fNnpqlep_9M6Prbp8xqs7K6q8" data-hostname="4f8b16e0ad974eb7b7d929125dbdc1f.svc.dynamics.com">
29 </div> <!-- End MS Dynamics --> <!-- Google Tag Manager --> <script>
30 (function(w, d, s, l, i) {
31   w[l] = w[l] || []; w[l].push((
32     'gtm.start':
33     new Date().getTime(), event: 'gtm.js'
34   )); var f = d.getElementsByTagName(s)[0],
35   j = d.createElement(s), dl = l != 'dataLayer' ? '&l=1' : ''; j.async = true; j.src =
36   'https://www.googletagmanager.com/gtm.js?id=' + i + dl; f.parentNode.insertBefore(j, f);
37   })(window, document, 'script', 'dataLayer', 'GTM-M53PKPD');</script> <!-- End Google Tag Manager --> <!-- Global
38 site tag (gtag.js) - Google Analytics --> <script async src="https://www.googletagmanager.com/gtag/js?id=UA-
39 113138049-1"></script> <script>
40 window.dataLayer = window.dataLayer || [];
41 function gtag() { dataLayer.push(arguments); }
42 gtag('js', new Date());
43 gtag('config', 'UA-113138049-1', {
44   send_page_view: false
45 });
46 </script> <!-- Google Tag Manager (noscript) --> <noscript> <iframe src="https://www.googletagmanager.com
47 /ns.html?id=GTM-M53PKPD"
48 height="0" width="0" style="display:none;visibility:hidden"></iframe> </noscript> <!-- End Google Tag
49 Manager (noscript) -->
```

JavaScript and Python (Slide 3 of 3)

The image shows a Python IDE with three windows:

- test.py**: A Python script with the following code:

```
def fullname(name,surname):  
    return name + " " + surname  
#This ends the function definition; the next line calls the function  
greeting = 'Hello ' + fullname('World', '')  
name = 'World'  
surname = ''  
if name == 'World':  
    print('Enter your first name')  
    name = input()  
    print('Enter your surname')  
    surname = input()  
#This ends the loop; the next line is not indented  
greeting = 'Hello ' + fullname(name,surname)  
print(greeting)
```
- *Python 3.7.2 Shell***: A command prompt window showing the execution of the script:

```
Python 3.7.2 (tags/v3.7.2:9a3ffc0492, Dec 23 2018, 21:52:03) [Intel] on win32  
Type "help", "copyright", "credits" or "license()" >>>  
[DEBUG ON]  
>>>  
= RESTART: C:\Users\James\AppData\Local\Programs\Python\Python37-32\python.exe test.py  
Enter your first name
```
- Debug Control**: A window for debugging the script. It shows the current line of execution as `> '_main_.<module>(), line 9: name = input()'`. The **Locals** pane shows the current state of the program's variables:

Locals	
<code>__annotations__</code>	<code>{}</code>
<code>__builtins__</code>	<code><module 'builtins' (built-in)></code>
<code>__doc__</code>	<code>None</code>
<code>__file__</code>	<code>'C:\\Users\\James\\AppData\\Local\\Programs\\Python\\Python37-32\\test.py'</code>
<code>__loader__</code>	<code><class 'frozen_importlib.BuiltinImporter'></code>
<code>__name__</code>	<code>'_main_'</code>
<code>__package__</code>	<code>None</code>
<code>__spec__</code>	<code>None</code>
<code>fullname</code>	<code><function fullname at 0x00F0C618></code>
<code>greeting</code>	<code>'Hello World '</code>
<code>name</code>	<code>'World'</code>
<code>surname</code>	<code>''</code>

Use Cases for Scripting

- API use
- Restarting machines
- Remapping network drives
- Installation of applications
- Updating system OS and apps
- Automated backups
- Information gathering
 - Inventory

Scripting Best Practices and Considerations

- Malware risk
- Inadvertent system/settings changes
- Browser or system crash due to resource mishandling

Lab Activity

- Lab: JavaScript
(available in CertMaster Learn and CertMaster Perform)
- Live Lab: Implement a PowerShell Script
(available in CertMaster Perform and CertMaster Labs)
- Live Lab: Implement a Bash Script
(available in CertMaster Perform and CertMaster Labs)

Lesson Review: Scripting Basics

- Shell Scripts
- Basic Script Constructs
- Windows Scripts
- JavaScript and Python
- Use Cases for Scripting
- Scripting Best Practices and Considerations